

FYS5419/9419 January 28



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1 qubit basis

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

2 qubit basis

$$|0\rangle_A \otimes |0\rangle_B = |00\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|0\rangle_A \otimes |1\rangle_B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = |01\rangle$$

$$|10\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}_A \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix}_B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} =$$

$$|1\rangle_A \otimes |0\rangle_B$$

$$|11\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |1\rangle_A \otimes |1\rangle_B$$

Pauli matrices

$$\sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Z \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = +1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Z \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\hat{V}_X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = X$$

$$X \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X |0\rangle = |1\rangle$$

$$\hat{V}_Y = Y = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

Block sphere:

$$|4\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$\alpha = r_0 e^{i\varphi_0}$$

$$\beta = r_1 e^{i\varphi_1}$$

$$|\psi\rangle = r_0 |0\rangle + r_1 e^{i(\varphi_1 - \varphi_0)} |1\rangle$$

$$\varphi = \varphi_1 - \varphi_0$$

$$0 \leq \varphi \leq 2\pi \quad r_0^2 + r_1^2 = 1$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + \sin\left(\frac{\theta}{2}\right) e^{i\varphi} |1\rangle$$

$$0 \leq \theta \leq \pi$$

1-qubit basis -

$$|4\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\mathbb{1} = |0\rangle\langle 0| + |1\rangle\langle 1|$$

$$P(0) = |0\rangle\langle 0| = P$$

$$P^2 = P \quad (\text{idempotent})$$

$$Q^2 = Q = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathbb{1} = P + Q$$

$$P|4\rangle = \alpha|0\rangle \wedge Q|4\rangle = \beta|1\rangle$$

Measurements

$$\hat{A}|\psi_a\rangle = \lambda_a|\psi_a\rangle$$

Projection operators-

$$\{\hat{P}_0, \hat{P}_1, \dots, \hat{P}_{n-1}\}$$

$$\mathbb{I} = \sum_{i=0}^{n-1} \hat{P}_i$$

Example: $|0\rangle$ and $|1\rangle$

$$|0\rangle\langle 0| = \hat{P}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$P_{\text{prob}}(\lambda_i) = |\hat{P}_n |\psi_a\rangle|^2$$

$$= \langle \psi_a | \hat{P}_n^\dagger \hat{P}_i | \psi_a \rangle$$

$$\left\{ \begin{array}{l} (|\psi_a\rangle = \sum_{i=0}^{n-1} a_i |i\rangle) \\ \hat{P}_i^\dagger \hat{P}_n = \hat{P}_n^2 = \hat{P}_i \end{array} \right\}$$

$$= \langle \psi_a | \hat{P}_n | \psi_a \rangle$$

$$\text{Total Prob: } \sum_{i=0}^{n-1} P_{\text{prob}}(\lambda_i) = \underline{1}$$

After measurement, the
normalized state

$$\frac{\hat{P}_a |\psi\rangle}{\sqrt{\langle \psi | \hat{P}_a | \psi \rangle}}$$

Example: 1-qubit base

$$\hat{P}_0 = |0\rangle\langle 0| \quad \hat{P}_1 = |1\rangle\langle 1|$$
$$\sum \hat{P}_i^\dagger \hat{P}_i = \underline{1}$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$P_{\text{prob}}(\psi_0) = \langle\psi| \hat{P}_0^\dagger \hat{P}_0 |\psi\rangle$$

$$= (\alpha^* \langle 0| + \beta^* \langle 1|) |0\rangle \langle 0| \\ \times (\alpha|0\rangle + \beta|1\rangle)$$

$$= |\alpha|^2$$

$$P_{\text{prob}}(\psi_1) = |\beta|^2$$

$$|4_0'\rangle = \frac{\hat{P}_0 |4\rangle}{\sqrt{\langle 4 | \hat{P}_0 | 4 \rangle}} =$$

$$\frac{\alpha |c\rangle}{|\alpha|}$$

$$|4_1'\rangle = \frac{\beta}{|\beta|} |1\rangle$$

Density matrix / operator

$$\hat{\rho}_4 = |4\rangle\langle 4|$$

$$|4\rangle = \alpha|c\rangle + \beta|1\rangle$$

$$|4\rangle\langle 4| = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$$

$$\hat{\rho}_c + \hat{\rho}_0 = \begin{bmatrix} 1 & c \\ 0 & 0 \end{bmatrix} = \hat{\rho}_c$$

$$\hat{\rho}_c \hat{\rho}_4 = \begin{bmatrix} 1 & 0 \\ c & 0 \end{bmatrix} \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$$

$$= \begin{bmatrix} |\alpha|^2 & \alpha \beta^* \\ 0 & 0 \end{bmatrix}.$$

$$P_{\text{rob}}(\psi_0) = |\alpha|^2 =$$

$$\text{Tr} \left[\hat{P}_0^\dagger \hat{P}_0 \hat{\rho}_\psi \right] = |\alpha|^2$$

in general

$$P_{\text{rob}}(x) = \text{Tr} \left[\underbrace{\hat{P}_x^\dagger \hat{P}_x}_{\hat{M}_x} \hat{\rho} \right]$$

Density matrix

$$|4\rangle = \sum_i c_i |i\rangle$$

$$c_i = \langle 4 | i \rangle$$

$$\hat{\rho}_4 = |4\rangle \langle 4| = \sum_i |c_i|^2 \underbrace{|i\rangle \langle i|}_{P_i}$$

$$P_i = |c_i|^2$$

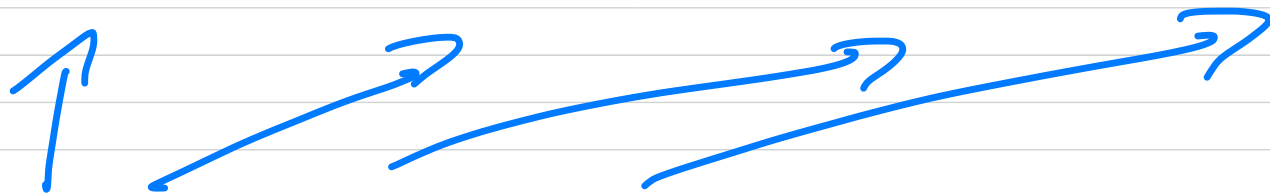
spectral decomposition

$$\hat{\rho}_4 = \sum_i P_i |i\rangle \langle i| = \sum_i P_i \hat{P}_i$$

Bell states (density matrices
entanglement,
pure and mixed
states + t)

Two-qubit bases

$|00\rangle, |01\rangle, |10\rangle, |11\rangle$



These all product states

$$|00\rangle = |0\rangle \otimes |0\rangle$$

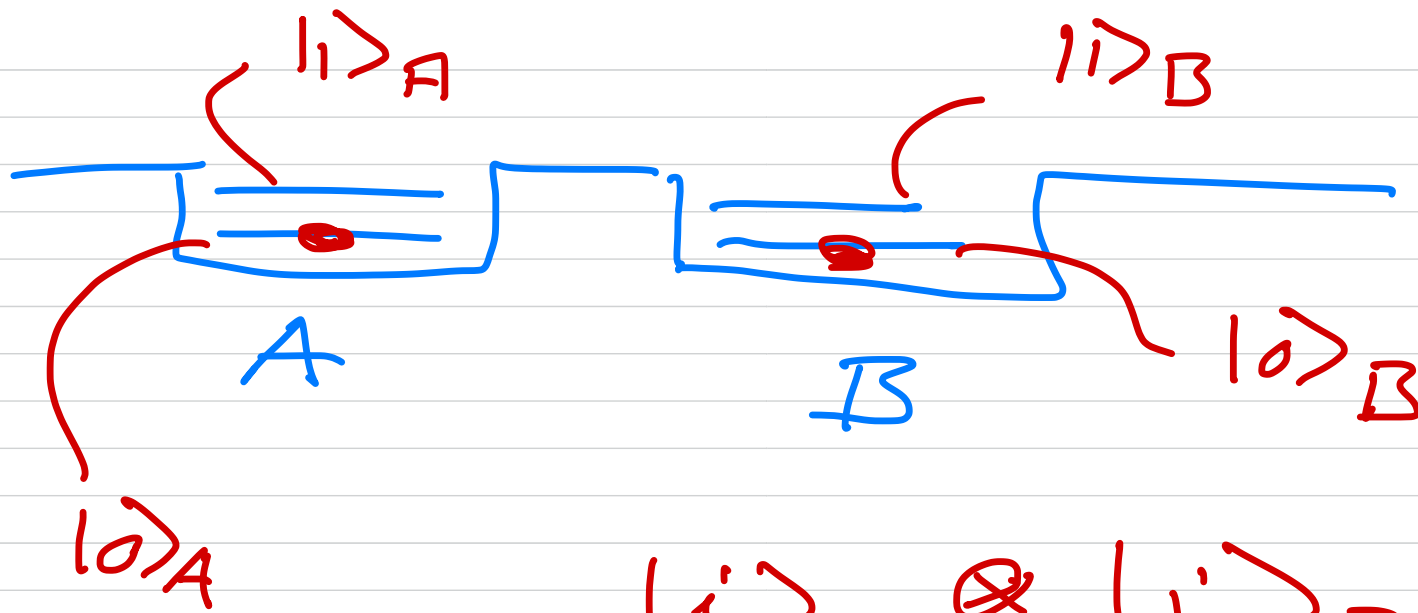
$$|\phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle + |11\rangle]$$

entangled states because they cannot be written as a product of the comp.

basis states $|00\rangle, |10\rangle, |01\rangle$
and $|11\rangle$

Example : atomic helium

$$\psi(x_1, x_2) = \varphi_{1s}(\vec{r}_1) \varphi_{1s}(\vec{r}_2) \\ \times \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$



$$|i\rangle_A \otimes |j\rangle_B$$

$$i = 0, 1$$

ground state

$$|0\rangle_A \otimes |0\rangle_B$$

excited state

$$\propto |00\rangle + \beta |11\rangle$$

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \wedge |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} [|00\rangle - |11\rangle]$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}} [|10\rangle + |01\rangle]$$

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} [|10\rangle - |01\rangle]$$

$$\rho_{\phi^+} = \frac{1}{2} [|00\rangle + |11\rangle] [\langle 00| + \langle 11|]$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\rho_{\phi^+}^2 = \rho_{\phi^+}$$

$$\text{Tr} [\rho_{\phi^+}^2] = \underline{1}$$

Measurements:

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B$$

↑

$$P_0 = |0\rangle_A \langle 0|_A \otimes I_B$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{2 \times 2}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Prob(\phi^+_{(A)}) = \quad \left(\begin{bmatrix} 1 \\ c \\ c \\ c \end{bmatrix} \right)$$

$$\langle \phi^+ | \hat{P}_0 | \phi^+ \rangle_{AB} \quad (11)$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & c \\ 0 & 1 & 0 & c \\ c & 0 & c & 0 \\ c & 0 & c & 0 \end{bmatrix} \begin{bmatrix} 1 \\ c \\ c \\ c \end{bmatrix}$$

$$= \frac{1}{2}$$

$$Prob(\phi^+_{(A)}) = 1/2$$